

CS308 Java Programming - Syllabus

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Course Information

- Course code: CS308
- Credits: 3
- Semester: 2025-26 EVEN
- Prerequisites: CS101 Programming Fundamentals

Course Objectives

- Write object-oriented programs in Java.
- Use collections, generics, exceptions and streams effectively.
- Build concurrent and database-backed Java applications.

Unit 1: Java Fundamentals

Unit 1 of CS308 covers JVM, JRE and JDK, data types and operators, control statements, arrays and strings and methods and overloading.

Unit 2: Object-Oriented Programming

Unit 2 of CS308 covers classes and objects, inheritance and polymorphism, abstract classes and interfaces, packages and access modifiers and exception handling.

Unit 3: Collections and Generics

Unit 3 of CS308 covers List, Set and Map, iterators, generics, Comparable and Comparator and streams and lambda expressions.

Unit 4: Concurrency

Unit 4 of CS308 covers threads and Runnable, synchronization, executor services, concurrent collections and CompletableFuture.

Unit 5: Java Applications

Unit 5 of CS308 covers JDBC, file I/O, unit testing with JUnit, build tools: Maven and introduction to Spring Boot.

Course Outcomes

- CO1: Write Java programs using OOP principles.
- CO2: Use the collections framework and generics.
- CO3: Handle exceptions and write unit tests.
- CO4: Write thread-safe concurrent code.
- CO5: Connect Java applications to a database with JDBC.

Assessment

CS308 is assessed out of 100 marks: 40 internal marks and 60 external marks. Internal marks consist of two Continuous Assessment Tests (CAT 1 and CAT 2) of 15 marks each and assignments worth 10 marks. The end-semester examination is conducted for 100 marks and scaled to 60. Lab work carries 20 of the 40 internal marks, and the CAT weight is halved.

Textbook

Herbert Schildt, Java: The Complete Reference, 12th edition.